

the protocol: a Python class in the labpack (abridged)

```
class AudiovisualPairing(BaseProtocol):  
    """A pulse song and a moving patch, started together."""  
  
    def get_protocol_parameter_defaults(self):  
        return {'pre_time': 0.5,  
                'stim_time': 2.0,  
                'tail_time': 1.0,  
                'freq': [225.0, 120.0],  
                'intensity': 0.0,  
                'center': (0, 0),  
                'speed': 60.0,  
                'angle': 0.0,  
                'volume': 0.5}  
  
    def get_trial_parameters(self):  
        ... # each trial's MovingPatch + PulseSong
```

```
# each trial, stimpack turns the descriptors into module calls:  
server.target('visual').load_stim(name='MovingPatch', theta=..., ...)  
server.target('audio').load_stim(name='PulseSong', freq=225.0, ...)  
sleep(pre_time) # gray screen, silence  
server.target('all').start_stim() # one start, both modalities  
sleep(stim_time) # the patch sweeps, the song plays  
server.target('all').stop_stim()  
sleep(tail_time)
```

the GUI's Main tab, built from those declarations, mid-run

The screenshot shows the 'Main' tab of a GUI. At the top, there are tabs for 'Main', 'Ensemble', 'Subject', and 'File'. Below these, there are dropdown menus for 'Protocol' (set to 'AudiovisualPairing') and 'Param preset' (set to 'Default'), along with 'Save preset' and 'Delete preset' buttons. The 'Run parameters' section includes input fields for 'num_trials' (40), 'idle_color' (0.5), and checkboxes for 'all_combinations' and 'randomize_order' (both checked). The 'Protocol parameters' section includes input fields for 'pre_time' (0.5), 'stim_time' (2.0), 'tail_time' (1.0), 'freq' ([225.0, 120.0]), 'intensity' (0.0), 'center' ((0, 0)), and 'speed' (60.0). A red box highlights the 'freq' field, and a red arrow points from it to the 'freq' field in the 'Run parameters' section. Below the 'Protocol parameters' section, there is a 'This trial: ffreq: 225.0' label. At the bottom, there are input fields for 'Series #' (1) and 'Subject' (subject-1), and a progress bar showing 'Elapsed / Est: 8 / 140s' and 'Trials run: 2 / 40'. There are buttons for 'View', 'Record', 'Pause', and 'Stop'. At the very bottom, there is a 'Recording series 1' label and a 'Note' button.

Main Ensemble Subject File

Protocol: AudiovisualPairing

Param preset: Default Save preset Delete preset

Run parameters:

num_trials: 40

idle_color: 0.5

all_combinations: ☒

randomize_order: ☒

Protocol parameters:

pre_time: 0.5

stim_time: 2.0

tail_time: 1.0

freq: [225.0, 120.0]

intensity: 0.0

center: (0, 0)

speed: 60.0

This trial: ffreq: 225.0

Series #: 1 Subject: subject-1

Elapsed / Est: 8 / 140s Trials run: 2 / 40

View Record Pause Stop

Recording series 1 Note